

JY-CRPG-BENCH: A LONG-HORIZON BENCHMARK FOR VISION-LANGUAGE MODELS IN A CHINESE OPEN-WORLD RPG

Han Xiao

han.xiao@jina.ai

Yiming Liu

letusgo126@gmail.com

ABSTRACT

A long-horizon task requires a model to perceive its environment, decide for itself what is worth doing, and hold one plan across hours of dependent decisions. We present JY-CRPG-BENCH, a benchmark that places a vision-language model inside 金庸群俠傳, the 1996 open-world role-playing game *Heroes of Jin Yong*, run under DOS emulation, and scores it against the ending the game defines, which takes a first-time human player tens of hours. The model sees a 320×200 frame and presses keys. The game speaks only Traditional Chinese, draws its world in an isometric projection in which no key moves straight up the screen, and states its objectives only inside its fiction. Progress is read from the memory of the emulated machine and from the save the game writes, so a ladder of state-verified milestones orders the field until completion time can. Under a 60-minute budget, all 12 frontier models reach the world map, one holds the compass, and none gains experience. The benchmark, its replays and every session are public at <https://github.com/hanxiao/jy-crpg-bench>.

1 INTRODUCTION

Frontier models prove competition mathematics and write working compilers, yet the tasks they complete reliably are still measured in hours of human work (Kwa et al., 2025). A long horizon is a property of the task, the number of dependent steps its completion requires (Chen et al., 2026), and its evaluation takes two forms. Episodic benchmarks set a goal and score its completion; continuing benchmarks give the model an environment with no terminal state and score what accumulates (Backlund & Petersson, 2025; Fan et al., 2026). An open-world game offers both, an ending that takes tens of hours to reach and a record of everything that accumulates before it, and adds a requirement the business simulations do not: the model has to recover its own state from raw perception, because nothing in the environment reports it.

Existing game benchmarks each remove one part of that problem to isolate another. BALROG evaluates six research environments with a language-only mode (Paglieri et al., 2025), TextQuests removes perception entirely (Phan et al., 2025), ARC-AGI-3 excludes language and cultural priors by construction (ARC Prize Foundation, 2026), VideoGameBench detects progress by matching the screen against walkthrough checkpoints (Zhang et al., 2025a), and MineExplorer filters out tasks whose solutions depend on game-specific knowledge (Ju et al., 2026). Each removal is principled, and each leaves the combination untested. We keep the parts together, because acting through unfamiliar perception, unfamiliar language and unfamiliar culture at once is what a deployed model is asked to do.

JY-CRPG-BENCH runs a single deep environment, following the precedent of Crafter and NetHack (Hafner, 2021; Küttler et al., 2020). The environment is 金庸群俠傳, a 1996 Chinese role-playing game that remains a landmark of its genre, executed as its original DOS binary under emulation. The game presents an open world built from the fourteen novels of Jin Yong: a two-tier map, several hundred named characters each carrying level, experience, skills, items and reputation, and an ending that requires recovering fourteen books scattered across the world. A first human playthrough takes tens of hours. The model receives the framebuffer and a key vocabulary through one fixed, minimal harness, and nothing else, so what varies between runs is the model.



Figure 1: The environment at the resolution the model receives it, upscaled three times without interpolation: the two tiers of the world (a, b), the four axes the movement keys run along (c), the three kinds of text panel through which every objective, choice and statistic is stated (d–f), the bag (g), the outdoor menu with the party list it opens (h), and a turn of battle with the menu and record of the acting character (i).

We measure against that ending. The fourteen books sit behind items, factions and martial arts that have to be acquired in order, in a world that reports nothing about itself, and the game records whether they have been recovered. A model that finished would have held one plan across tens of hours of self-directed play, which is the capability long-horizon evaluation exists to detect. The instrument therefore spans both phases of play: a ladder of state-verified milestones orders the field while models are learning to move through the world, and completion time orders it once they finish, so the benchmark gains a speedrun ranking where a pass-or-fail score would saturate.

Three properties make the environment hard for current models, and Figure 1 shows the screens they apply to. First, perception is difficult on several documented axes at once. Downsampling costs vision-language models accuracy (Niu et al., 2025), degraded input costs them more (Usama et al., 2025), out-of-distribution images degrade recognition even of familiar categories (Lin et al., 2026), text rendered as pixels is understood less well than the same text supplied as tokens (Liu et al., 2026), and precise spatial reading fails once primitives are small or adjacent (Rahmanzadehgervi et al., 2024; Wang et al., 2024); a 320×200 frame of 1996 pixel art carrying 16-pixel Traditional Chinese glyphs sits at the hard end of all five. Second, the world is drawn in a 45° isometric projection, so the four movement keys run along screen diagonals, and a model that reasons in screen coordinates walks in circles. Third, progress is self-directed. The game keeps no quest log, draws no objective marker and reports no score. The brief supplies the opening route, from the first room to the compass, and a manual of the controls; from the compass on, deciding where to go and what is worth doing is part of the problem, the goal-setting requirement that ARC-AGI-3 places at the centre of agentic intelligence (ARC Prize Foundation, 2026).

Measuring such an environment from the screen is unreliable, and recent work says so: GameWorld names heuristic verification as the obstacle to standardised game-agent evaluation and pairs each of its tasks with a state-verifiable metric (Ouyang et al., 2026). We take the same position inside the

Table 1: Interactive agent benchmarks. *Observation* is what the model receives before any upscaling, *self-directed* marks environments that supply no task list, score or objective marker during play, and *progress from* is what a progress claim is checked against.

Benchmark	Environment	Observation	Language	Self-directed	Progress from
BALROG	6 research games	text, render	English	no	env. reward
TextQuests	interactive fiction	text	English	yes	game state
ARC-AGI-3	abstract puzzles	64×64 grid	none	yes	env. score
VideoGameBench	1990s action games	native px	English	no	checkpoint images
GameWorld	34 browser games	browser	English	no	game state
MineExplorer	Minecraft	3D render	English	no	milestone rules
JY-CRPG-BENCH (ours)	one 1996 CRPG	320×200 px	Trad. Chinese	yes	memory, save file

emulator. The benchmark reads progress from the memory of the running machine, where the game maintains its character array, and from the save file the game writes when asked, which holds the party roster and the world position. The model has no call that returns any of this, and nothing read from the screen enters a score.

We make three contributions.

1. We present JY-CRPG-BENCH, a long-horizon benchmark on a commercial open-world role-playing game whose difficulty comes from named and independently checkable properties of the environment: low-resolution perception of 1996 pixel art, isometric control, Traditional Chinese grounding, a keyboard action space of which the game answers a small subset, and persistent character state that only coherent play advances.
2. We score progress in the terms the game keeps. Character records, the party and the fourteen books are read from emulator memory and from a save the game writes for itself, validated against the archives the release ships, and measurement never writes to the machine it observes. Every rung of the ladder is an event the model has to cause, and the terminal rung is the ending the game defines.
3. We evaluate 12 frontier models under the same agent harness and against a seeded random baseline under a wall-clock budget, and release the benchmark and every recorded session as a public, replayable leaderboard.

Table 1 places JY-CRPG-BENCH among interactive agent benchmarks. The next section reviews game environments as agent benchmarks and the measurement problem that long horizons create, Section 3 details the environment, the interface, the session protocol and the measurement, and Section 4 reports the field: how much of the ladder is open, how far it separates models, how the budget is spent, and what the replays show.

2 RELATED WORK

2.1 GAME ENVIRONMENTS AS AGENT BENCHMARKS

Games have been the standard setting for sequential decision-making since the Arcade Learning Environment (Bellemare et al., 2013), and language and vision-language agents have produced a second generation of game benchmarks. BALROG aggregates six reinforcement-learning environments under one protocol and reports a persistent gap between what a model knows about a game and what it does inside one (Paglieri et al., 2025). Igame-Bench isolates how much of a game result belongs to the harness (Hu et al., 2026), and Orak spans twelve popular titles with configurable agentic modules (Park et al., 2026). VideoGameBench is the nearest benchmark to ours, running real 1990s games from raw pixels under a no-scaffold rule (Zhang et al., 2025a). GameWorld standardises the action interface across 34 browser games and pairs every task with a state-verifiable metric (Ouyang et al., 2026). MineExplorer evaluates open-world exploration in Minecraft and finds that models handle single-hop tasks but degrade once hidden prerequisites must be coordinated over a longer trajectory (Ju et al., 2026), which is the structure of the opening sequence in our environment as well. The nearest precedents in horizon are the public Pokémon runs. Claude 3.7 Sonnet reached

three gym badges from screen pixels behind a memory scaffold, where an earlier model had not left the first house (Anthropic, 2025). Gemini 2.5 Pro finished Pokémon Blue in 813 hours behind a harness that translated the RAM of the game into text, and an ablation without vision performed about as well (Comanici et al., 2025). The PokeAgent Challenge turned the setting into a speedrunning track scored by completion time over milestones (Karten et al., 2026a). PokeGym builds a long-horizon benchmark on the open-world game Pokémon Legends: Z-A with no access to game state (Zhang et al., 2026a), and StarBench one on the turn-based Honkai: Star Rail (Zhang et al., 2025b). Crafter established the single-environment, achievement-based design we follow (Hafner, 2021), and NetHack remains the deepest environment in use for learned agents (Küttler et al., 2020). JY-CRPG-BENCH differs from these in running one persistent open world with Traditional Chinese as its only text, in verifying progress from emulator memory and from the save file of the game, and in scoring against the completion condition the game defines.

2.2 LONG-HORIZON EVALUATION AND ITS MEASUREMENT

Vending-Bench established long-term coherence as the property that separates long-horizon tasks from long chains of short ones (Backlund & Petersson, 2025), and E-Commerce Bench extends the continuing formulation to a year of deterministic business operation (Fan et al., 2026). Both keep perception out of the loop. TextQuests shares our self-directed, long-horizon framing while removing vision (Phan et al., 2025), TextAtari stretches horizons to 10^5 steps over text renderings (Li et al., 2025), ARC-AGI-3 isolates adaptation to novel tasks by excluding language and external knowledge (ARC Prize Foundation, 2026), and AutumnBench probes world-model quality through environment-level queries (Warrier et al., 2025). UltraHorizon makes exploration under hidden rules the task itself (Luo et al., 2025), and Kapoor et al. (2026) observe that benchmark design has favoured tasks that are precisely specified, automatically graded and short, and call for open-world evaluation. Navigation from rendered images is a documented weakness: on mazes, one-shot navigation collapses for every model by ten cells a side (Jiang et al., 2026), and the failures are attributed to looping (Einarsson, 2025).

Measurement is where long horizons strain evaluation. The ABC checklist catalogues task-validity and outcome-validity failures across agentic benchmarks (Zhu et al., 2025), an audit of agent traces finds reward hacking in a majority of them on two suites (Shao et al., 2026), and a survey of the horizon gap finds that the field answers uninformative outcome signals by building denser step-level ones (Chen et al., 2026). State-based verification has precedent outside games, where τ -bench compares the final database state with an annotated goal (Yao et al., 2024). Our milestone ladder takes the same approach, read from machine state so that dense signals cannot be mistaken for progress.

3 JY-CRPG-BENCH

3.1 ENVIRONMENT

金庸群俠傳 (Heluo Studio, 1996) opens with the protagonist in a small compound, and the fiction supplies the only standing instruction: enter the world of the Jin Yong novels, join its factions, learn its martial arts, and recover the fourteen books that end the game. The world is two-tier, with many local scenes such as buildings, sects and caves attached to one outdoor map of 480×480 tiles (scarsty & contributors, 2026). For each of 320 named characters the game maintains a record carrying level, experience, health, stamina, learned skills, carried items, reputation and aptitude. Progression is nonlinear and gated. Every scene carries an entry condition, and at the start all of them are closed except the home, five inns and the house of the hermit 南賢; events open the rest, and the opening sets their order.

The compound holds a searchable chest, one speaking character and one exit tile. Beyond it, the path south leads to the hermit, whose long conversation opens the scenes the game keeps closed until then, the house of the first companion among them, and a cabinet beside him holds the compass, the only source of coordinates the game offers. The companion joins after a yes-or-no prompt at which the confirm key means no. Fights are turn-based on the scene grid. Each turn a character moves along the four axes and then strikes, uses an item, waits or rests. Skills draw on inner force, and the first scripted fight opens by draining it. An automatic mode hands the character to the AI

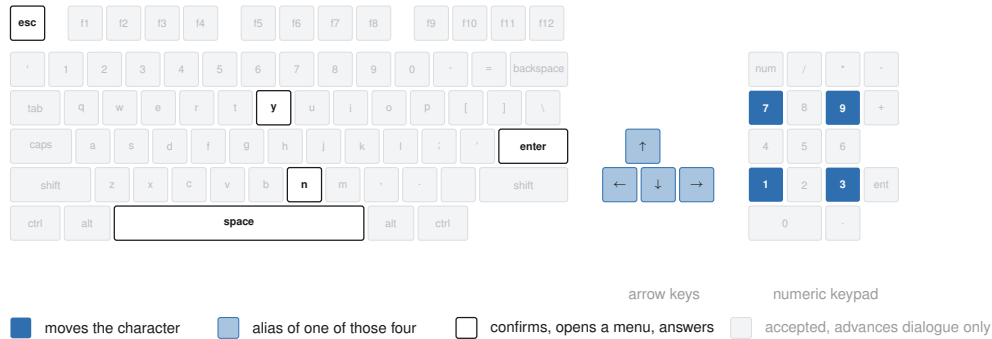


Figure 2: The keyboard the API exposes. The nine keys the game answers outside dialogue are picked out, the arrow keys are aliases of the four movement keys, and any key advances a dialogue.

of the game, which alternates striking and resting and loses to an opponent who heals. A party that wanders elsewhere first meets storylines whose fights it cannot survive at level one, and a lost fight ends the game, which then offers nothing but its own load menu.

Every objective, prompt and statistic reaches the model as Traditional Chinese rendered at sixteen pixels a line, and the character sheet on screen displays the same record the benchmark reads from memory. We selected this title for four reasons. Its text is Traditional Chinese throughout, a script that multimodal evaluation has largely neglected (Tam et al., 2025), so cross-lingual grounding is a central obstacle. It is menu-driven, so a model that deliberates between actions is not structurally disadvantaged as it would be in a real-time title (cf. Zhang et al., 2025a). Its progression state is rich enough to measure at the level of individual character statistics. Its 45° isometric projection makes the mapping from screen to action something the model must infer.

3.2 OBSERVATION AND ACTION SPACE

The model observes the raw framebuffer at 320×200 and acts by pressing keys. A scored session serves the native frame without upscaling, annotation, set-of-marks overlay or accessibility tree (cf. Xie et al., 2024). The action space is the DOS keyboard, with 130 key names over 100 distinct keys, and Figure 2 shows how few of them the game answers. Movement runs along the four numpad diagonals, with the arrow keys aliased onto the same axes, so a model that presses up expecting to walk up the screen moves up and to the right. The view follows the character except near the edge of a scene, where the background holds still and the sprite walks across it, so a model that reads its movement from the scrolling of the background loses count of the tiles it walked there. Canopies and roofs are drawn over the character, and the entrance of a building is a single tile on one face of its wall, so reaching a door means walking around the compound to the side the door is on.

The one action presses a key, or a list of keys in order, holding each for a requested number of emulated frames, ten by default, and returns once the screen has reacted and held still, waiting through a scene transition if one starts, so the frame a model receives is the one to act on. There is no call that advances the game without input, since the game only moves on input, and the hold is counted in frames because the game samples its keyboard once per game-loop iteration. A scored session answers the first four calls of Table 2: the screen, the brief, the key names and the action. Three further calls save, list and restore emulator snapshots for development, and a scored session refuses them, so a run cannot be rewound from outside the game. The save and load menu of the game stays open to the model as to a human player, so a model that loses a fight can reload the last save the game wrote, and the recording shows it doing so.

3.3 HARNESS AND SESSION PROTOCOL

The benchmark supplies the brief and the control surface, and the model brings an agent. Every model in this paper ran inside `pi`¹, a minimal coding agent installed as shipped, whose four tools read,

¹<https://github.com/badlogic/pi-mono>

Table 2: The control surface. A scored session answers the first four calls.

Call	Body	Meaning
GET /api/screen	–	the current frame
GET /api/help	–	the brief, in English or Chinese
GET /api/keys	–	every accepted key name
POST /api/key	{ "key": "kp3", "hold": 10 }	press one key, or a list of keys in order
GET /api/slots	–	emulator snapshots on disk
POST /api/save	{ "name": "... " }	take a snapshot
POST /api/load	{ "name": "... " }	restore a snapshot

write and edit files and run a shell, with no web access, so no model could look up a walkthrough; the model drives the control surface with HTTP calls from that shell. The brief is the whole instruction the model receives: the calls that constitute play, the rules of a run, the opening route as far as the compass, and a manual of the controls and systems of the game, served in English and Chinese with the on-screen Traditional Chinese terms kept verbatim, so the language a model reasons in is not itself a handicap. An action returns only its metadata, so the model looks at the screen when it decides to, and the benchmark adds no memory scaffold, no map, no pathfinder and no access to the state of the game; whatever a model keeps between actions it writes for itself in that shell. This is what makes the leaderboard a ranking of models: harness choice moves game results (Hu et al., 2026; Karten et al., 2026b), and on long-horizon tasks the variance a harness induces can exceed the variance between models (Zhang et al., 2026b).

Table 2 lists the control surface in full, and the brief is published as `agents.md`, the project file `pi` reads at launch. A scored session is refused every field that would report its own progress until the run ends, so a model cannot query its own score.

A session is created under a declared name with a playtime, 60 minutes by default and up to twenty-four hours, and the clock starts at the first playable moment. The wall clock of every decision call is kept on the server, so played time and think time are measured for any client and shown beside a run without ranking it. Each session is a separate emulator process with a private copy of the game, which every host supplies itself as described in Appendix B, with its save slots reset to a new game, so no run inherits the progress of another. A session ends only when its budget lapses or the game is completed, and there is no inactivity cutoff. When the budget lapses the session renders its recording to a time-compressed video, computes its metrics, and appends itself to the public catalogue with the video and its keypress timeline, so the behaviour behind a score can be reviewed. Figure 3 shows the apparatus, and Appendix A lists the emulation and hosting parameters.

3.4 STATE-VERIFIED PROGRESS

The game keeps its persistent state in one archive: a base block with the world position, the party roster and the shared bag, followed by 320 character records of 182 bytes and an item table. We validated that layout against the opening machine image and the archives the release ships, and we locate the same array in the live emulated machine by anchoring on one character name and confirming the two neighbouring records at the expected stride, since the name recurs in the image and the layout moves between runs. The bag in front of that array is the working copy and changes the moment an item is taken, so the item count, the compass and the books are read from it after every action and again from every save the game writes. A rung, once seen, is kept, so an item picked up and later spent still credits it. Reading is passive: the benchmark serialises the machine and never loads state into it. The only keys it presses on its own are those of the save described next, which take the game a few seconds of menu time that is not charged to the run and leave the screen as they found it.

For the party roster and the world position the benchmark reads the save the game writes. The game offers its save menu on the world map alone and announces this through the height of the menu, which stands at six rows on the world map and four in a scene, so an attempt opens the menu, reads its height, and saves only where six rows appear. An attempt recurs every two minutes and once more before the budget lapses, and waits for a two-second gap in the keys the model sends. The session is held while the attempt runs, so the model never sees the screen mid-save, and the time of

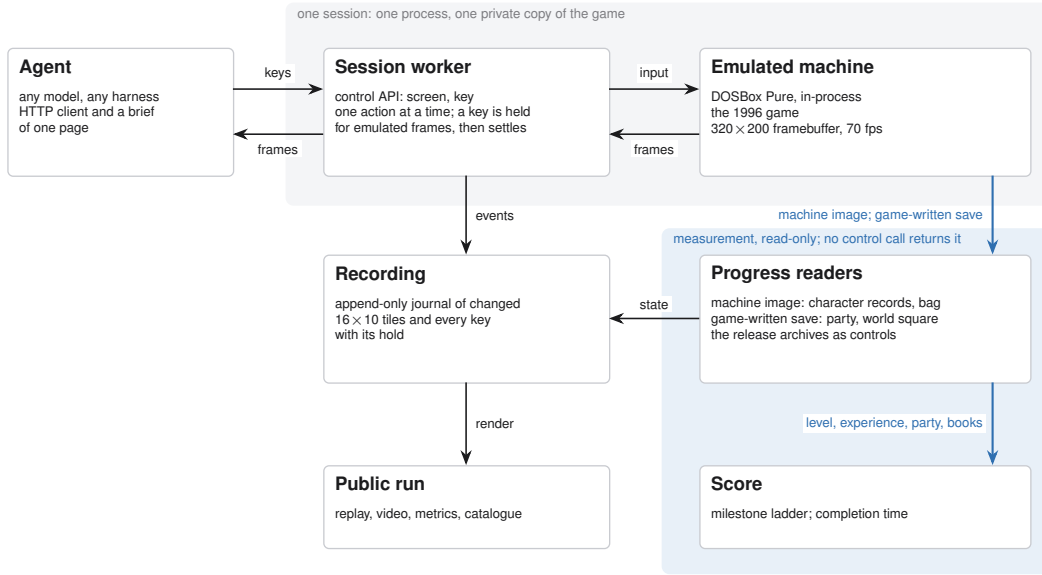


Figure 3: One process per session loads the game core and answers the control API; the recording feeds the public replay, and progress is read from the emulated machine and from the save the game writes, on a path no control call reaches.

Table 3: The milestone ladder. *Evidence* is the game state that credits the rung and *source* is where the benchmark reads it: the bag and character array in emulator memory, or the save the game writes on the world map.

Rung	Evidence	Source
<i>The opening, reached without a fight</i>		
picked something up	any item count above its opening count	bag in memory
reached the world map	the game wrote its save	save
opened the world	scenes closed at the start stand open	save
holds the compass	the compass in the bag	bag in memory
recruited a companion	a party roster longer than one	save
<i>The campaign, from the first fight on</i>		
gained experience	experience above zero	character record
reached level 2	level above one	character record
holds one of the fourteen	a book in the bag or carried by the protagonist	bag and record

the first save that lands is the time of the crossing. Neither the archive nor anything read from it is reachable through the control surface, so a save that lands is itself the evidence that the party has crossed onto the world map. The session record also keeps a screen fingerprint of the world map for the replay, and it never enters a score.

Progress is summarised as eight milestones read from this ground truth, in the spirit of the achievements of Crafter (Hafner, 2021), and Table 3 lists them with the evidence each rests on, every one a number the game keeps about itself. The rungs fall into two phases that the game itself separates. The opening needs no fight: the chest in the first room, the exit tile, the conversation of the hermit that opens the scenes the game keeps closed until then, the compass in his cabinet, and the companion whose house that conversation opens and who joins through dialogue are reached by walking, reading and answering. The campaign begins with the first fight, the first source of experience and levels, and runs through the chains of items, factions and fights that the books sit behind. The default budget is sized to the opening and the extended budgets to the campaign. The rungs are listed in the order the opening imposes, but a run is credited with every rung it reaches, so the count does not depend on the path taken.

The ladder runs to the ending of the game. The fourteen books, named in Appendix C, are what the game requires before its final sequence can be played, and the archive records every book held, so the count of books extends the ladder beyond its last rung, and the first read that finds all fourteen held records the completion time. A session that completes carries that time, which orders a field in which every entrant passes, so one instrument covers both phases and the benchmark keeps measuring after the first model finishes.

The ladder is the fixed score of the benchmark. For the multi-hour budgets the protocol allows, the release adds a second, additive report over the same ground truth: a finer trajectory of the scenes a run reached, the battles it survived, the skills it learned and the books it gathered, out to the ending, with the hard prerequisites of the walkthrough as dependency gates between them. Each component is scored against a fixed denominator with its coverage reported beside it, and the report never rescores the ladder, so a long run holding no book still leaves an ordered record of how far its plan carried. Its checkpoint contract and component weights are released with the benchmark.

3.5 RANDOM BASELINE

A seeded random policy plays the same budget through the same control surface, drawing from 13 keys the brief teaches, weighted towards movement and confirmation, and never reading the screen, following the convention that anchors a score on random and human play (Mnih et al., 2015; Hafner, 2021; ARC Prize Foundation, 2026). The rungs it reaches mark the progress available without reading the screen and are the floor of the ladder. Because it is seeded, its whole key sequence is recomputed offline and checked against the journal of keys the machine received.

4 EVALUATION

The field at submission is 12 models from 5 vendors at the 60-minute budget, and a seeded random baseline. Every session a model in the field played at this budget is on record, 42 in all, and a model is reported by the session that reached the most rungs, the most recent on a tie; every model played more than once, and Appendix D lists the sessions by model. Every session ran in pi under the same settings, and effort is reported as wall clock. Every session is public with its video and replay timeline, and the session records the paper is generated from are committed beside the paper source. The subsections that follow report how much of the ladder is open, how far it separates models, how the budget is spent, and what the replays show.

4.1 HEADROOM

Every rung is measured, and the highest rung any session reaches at this budget is the compass. Of the 12 sessions, 8 pick something up, and every session reaches the world map. One session, `gpt-6-astra`, completes the conversation of the hermit that opens the world and takes the compass from his cabinet, and it is the only session to clear a rung beyond the world map. No session recruits a companion, gains experience or holds a book: all 12 character records remain at level 1 with 0 experience and 1 skill, as they were when the game began, so every rung from the companion on is open. Figure 4 shows the ladder by model. Comparable benchmarks report their fields at a similar point: BALROG places its best model at 1.5% progression on NetHack (Paglieri et al., 2025), and VideoGameBench reports the best vision-language models completing 0.48% of its suite (Zhang et al., 2025a). The distance from the compass to the fourteen books is the headroom the benchmark holds.

4.2 DISCRIMINATION

The chest and the hermit divide the field, and they divide it differently. Every session crosses onto the world map, and the chest in the first room separates the 8 sessions that searched it from the 4 that crossed without it, so the count of rungs summarises what a run achieved and presumes no single path. The random policy reaches 1 of the eight rungs in 60 minutes: in a small room a random walk searches the chest within the hour. Its recomputed key sequence matches the journal of the machine key for key, and no save lands during its hour, so the world map is the first rung a random walk does not reach, and the chest rung separates models from one another but not from the floor. The

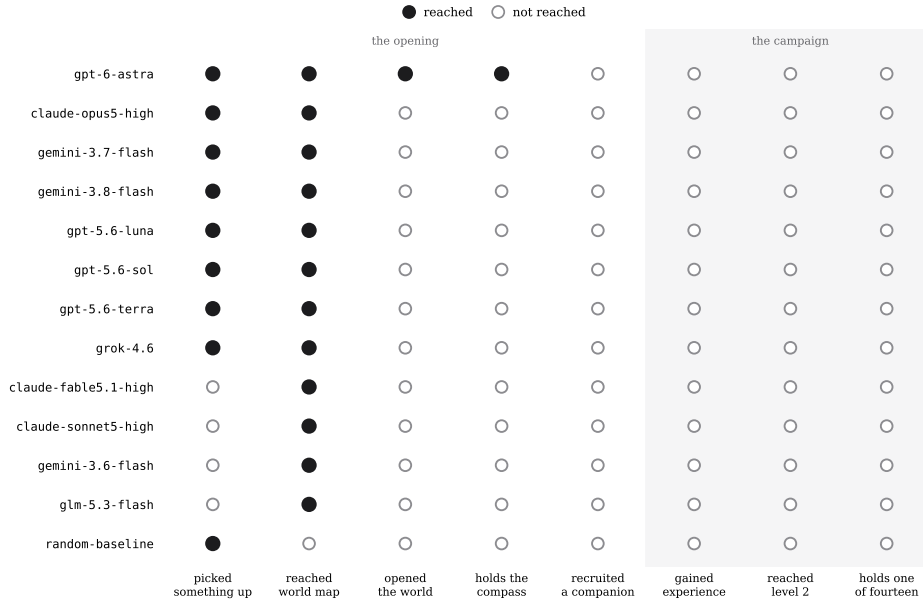


Figure 4: The milestone ladder, one row per model, with filled markers for rungs reached and hollow markers for rungs not reached.

hermit divides the field a second time, and above the compass the field is undivided, since no session recruits the companion. The world-map rung also shows why progress is read from the machine: of the 10 crossings that also carry a screen fingerprint for the replay, the save and the fingerprint agree on 9, the save credits 1 crossing the fingerprint missed, and the fingerprint never appears without a save behind it.

4.3 BUDGET

Of the 12 reported sessions, 11 played to their budget, and 1 ended early. The crossing onto the world map is timed by the game: the first save it writes there trails the crossing by at most the two-minute cadence of the attempts, and by that clock the 10 crossings whose save carries a time fall between 9 and 41 minutes into the session, with a median of 26. The session that went on to the compass crossed at 9 minutes, after 33 actions, and spent the rest of the hour beyond the exit, so the hour covers the first room and, for one session, the walk south to the cabinet. The budget is wall clock, so time spent deliberating is time not spent playing, an efficiency framing shared with ARC-AGI-3 and TextQuests (ARC Prize Foundation, 2026; Phan et al., 2025).

4.4 REPLAY ANALYSIS

The replays show where the hour goes, and Figure 5 collects one frame from each pattern. The session that holds the compass, *gpt-6-astra*, left the compound, walked the path south, and pressed through the conversation of the hermit in 7 minutes at 10 seconds a line before searching the cabinet. It then opened the item screen 9 times to read its coordinates from the compass, the one instrument the brief recommends, its 42 arrow-key presses all fell inside menus and prompts, never on the map, and it entered 11 scenes in the hour. Two other sessions of the same model, on record but not the one reported, went further than the ladder credits. One answered the yes-or-no prompt of the companion at an inn at minute 52 and was handed a skill book, the only recruitment in any replay; its record carries no party reading, so the ladder does not credit it. The other walked into the first scripted fight and sent no key after 128 actions, from minute 42 on. *gemin-3.8-flash* reached the conversation of the hermit and pressed through its 40 lines at 23 seconds each, so the conversation took 15 minutes and ended at minute 54; the hermit had named the cabinet, 6 minutes of budget remained, and the session ended without the compass.



Figure 5: Frames from the published replays, each with the strip of its video naming the model, the action count and the clock: (a) gpt-6-astra reads its coordinates from the compass on the item screen and (b) presses through the conversation of the hermit; (c) an earlier session of the same model at the prompt of the companion, and (d) another in the first scripted fight; (e) gemini-3.8-flash is told where the compass lies; (f) glm-5.3-flash in the yard behind the fence; (g) claude-fable5.1-high passing the gate of the hermit; (h) grok-4.6 talking to an innkeeper; (i) gpt-5.6-terra in the middle of a long key list.

The compound is left through a yard closed by a fence with one gate, and the yard held several sessions. glm-5.3-flash needed 1595 actions to cross, against a median of 130, and reached the world map at minute 37 after long stretches pressed into the fence. It is also the one session that never pressed enter or space, so the chest in the first room, which opens on either, stayed shut. claude-fable5.1-high crossed at minute 13 and then spent 1261 actions and 44 minutes on the world map without entering a scene, passing the gate of the hermit repeatedly, since the entrance of a walled compound is one tile on one face and every other face is solid. Two sessions that found an inn and a shop talked to the keeper repeatedly; the reply repeats, and nothing in it advances the opening.

gpt-5.6-terra sent lists of up to 100 keys in one action, 22 on average, with 32 actions of fifty keys or more, so it looked at the screen once for every 22 keys it pressed, and it reached the world map credited by the save alone.

5 CONCLUSION

We presented JY-CRPG-BENCH, a long-horizon benchmark that runs a 1996 open-world role-playing game, gives every model the same brief and control surface, and scores it in the terms the game keeps about itself. At the 60-minute budget the field of 12 models stays inside the opening of the game: every one of the 12 sessions reaches the world map, one holds the compass, and none gains

experience, so every rung from the companion to the fourteenth book stands open, and the one rung the random baseline reaches marks the floor a model has to clear. The environment is hard in ways that are named and checkable: the frame is a 320×200 isometric render of 1996 pixel art, objectives are stated only in Traditional Chinese, and the world supplies no score. Progress is read from the memory of the machine and from a save the game writes for itself, and every rung of the ladder is an event the model has to cause. The ladder runs to the ending of the game: the archive records the fourteen books held, so a model that finishes leaves proof of it, and from that point the leaderboard ranks by completion time, with the video and every key of the finishing run on record.

REFERENCES

- Anthropic. Claude’s extended thinking. Research post, section “Claude plays Pokémon”, <https://www.anthropic.com/research/visible-extended-thinking>, 2025.
- ARC Prize Foundation. ARC-AGI-3: A new challenge for frontier agentic intelligence. *arXiv preprint arXiv:2603.24621*, 2026.
- Axel Backlund and Lukas Petersson. Vending-Bench: A benchmark for long-term coherence of autonomous agents, 2025. URL <https://arxiv.org/abs/2502.15840>.
- Bahamut forum contributors. 原版金庸群俠傳攻略 (a complete walkthrough of the original game). <https://forum.gamer.com.tw/C.php?bsn=10859&snA=2778>, 2023.
- Marc G Bellemare, Yavar Naddaf, Joel Veness, and Michael Bowling. The arcade learning environment: An evaluation platform for general agents. *Journal of artificial intelligence research*, 47: 253–279, 2013.
- Mingguang Chen, Licheng Wang, and Bo Qu. The horizon gap: Planning, memory, execution, training, and evaluation for long-horizon LLM agents, 2026. URL <https://arxiv.org/abs/2608.06663>.
- Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities, 2025. URL <https://arxiv.org/abs/2507.06261>.
- Hafsteinn Einarsson. MazeEval: A benchmark for testing sequential decision-making in language models, 2025. URL <https://arxiv.org/abs/2507.20395>.
- Wei Fan, Xinjie Shen, Xudong Guo, Jianhong Tu, Yang Su, Yinger Zhang, Lianghao Deng, Fengyu Wang, Baohua Dong, Yangqiu Song, and Dayiheng Liu. E-Commerce Bench: Evaluating LLM agents on long-horizon autonomous business operation, 2026. URL <https://arxiv.org/abs/2608.30730>.
- Danijar Hafner. Benchmarking the spectrum of agent capabilities. *arXiv preprint arXiv:2109.06780*, 2021.
- Lanxiang Hu, Mingjia Huo, Yuxuan Zhang, Haoyang Yu, Eric P Xing, Ion Stoica, Tajana Rosing, Haojian Jin, and Hao Zhang. Imgame-Bench: How good are LLMs at playing games? In *International Conference on Learning Representations*, volume 2026, pp. 81308–81356, 2026.
- Yuhan Jiang, Peng Luo, and Liqiu Meng. Lost in aggregation: A multi-scale diagnostic benchmark for LLM spatial navigation, 2026. URL <https://arxiv.org/abs/2606.22219>.
- Tianjie Ju, Yueqing Sun, Zheng Wu, Wei Zhang, Yaqi Huo, Xi Su, Qi Gu, Xunliang Cai, Gongshen Liu, and Zhuosheng Zhang. MineExplorer: Evaluating open-world exploration of MLLM agents in Minecraft, 2026. URL <https://arxiv.org/abs/2605.30931>.
- Sayash Kapoor, Peter Kirgis, Andrew Schwartz, Stephan Rabanser, J. J. Allaire, Rishi Bommasani, Harry Coppock, Magda Dubois, Gillian K Hadfield, Andrew B. Hall, Sara Hooker, Seth Lazar, Steve Newman, Dimitris Papailiopoulos, Shoshannah Tekofsky, Helen Toner, Cozmin Ududec, and Arvind Narayanan. Open-world evaluations for measuring frontier AI capabilities, 2026. URL <https://arxiv.org/abs/2605.20520>.

- Seth Karten, Jake Grigsby, Tersoo Upaa Jr, Junik Bae, Seonghun Hong, Hyunyoung Jeong, Jaeyoon Jung, Kun Kerthaisong, Gyungbo Kim, Hyeokgi Kim, Yujin Kim, Eunju Kwon, Dongyu Liu, Patrick Mariglia, Sangyeon Park, Benedikt Schink, Xianwei Shi, Anthony Sistilli, Joseph Twin, Arian Urdu, Matin Urdu, Qiao Wang, Ling Wu, Wenli Zhang, Kunsheng Zhou, Stephanie Milani, Kiran Vodrahalli, Amy Zhang, Fei Fang, Yuke Zhu, and Chi Jin. The PokeAgent challenge: Competitive and long-context learning at scale, 2026a. URL <https://arxiv.org/abs/2603.15563>. NeurIPS 2025 Competition Track.
- Seth Karten, Joel Zhang, Tersoo Upaa Jr, Ruirong Feng, Wenzhe Li, Chengshuai Shi, Chi Jin, and Kiran Vodrahalli. Continual harness: Online adaptation for self-improving foundation agents, 2026b. URL <https://arxiv.org/abs/2605.09998>.
- Heinrich Küttler, Nantas Nardelli, Alexander Miller, Roberta Raileanu, Marco Selvatici, Edward Grefenstette, and Tim Rocktäschel. The NetHack learning environment. *Advances in Neural Information Processing Systems*, 33:7671–7684, 2020.
- Thomas Kwa, Ben West, Joel Becker, Amy Deng, Katharyn Garcia, Max Hasin, Sami Jawhar, Megan Kinniment, Nate Rush, Sydney Von Arx, Ryan Bloom, Thomas Broadley, Haoxing Du, Brian Goodrich, Nikola Jurkovic, Luke Harold Miles, Seraphina Nix, Tao Lin, Chris Painter, Neev Parikh, David Rein, Lucas Jun Koba Sato, Hjalmar Wijk, Daniel M. Ziegler, Elizabeth Barnes, and Lawrence Chan. Measuring AI ability to complete long software tasks, 2025. URL <https://arxiv.org/abs/2503.14499>. NeurIPS 2025.
- Wenhao Li, Wenwu Li, Chuyun Shen, Junjie Sheng, Zixiao Huang, Di Wu, Yun Hua, Wei Yin, Xiangfeng Wang, Hongyuan Zha, et al. TextAtari: 100k frames game playing with language agents. *arXiv preprint arXiv:2506.04098*, 2025.
- Ling Lin, Yang Bai, Heng Su, Congcong Zhu, Yaoxing Wang, Yang Zhou, Huazhu Fu, and Jingrun Chen. OODBench: Out-of-distribution benchmark for large vision-language models, 2026.
- Qing’an Liu, Juntong Feng, Yuhao Wang, Xinzhe Han, Yujie Cheng, Yue Zhu, Haiwen Diao, Yunzhi Zhuge, and Huchuan Lu. VISTA-Bench: Do vision-language models really understand visualized text as well as pure text?, 2026.
- Haotian Luo, Huaisong Zhang, Xuelin Zhang, Haoyu Wang, Zeyu Qin, Wenjie Lu, Guozheng Ma, Haiying He, Yingsha Xie, Qiyang Zhou, Zixuan Hu, Hongze Mi, Yibo Wang, Naiqiang Tan, Hong Chen, Yi R. Fung, Chun Yuan, and Li Shen. UltraHorizon: Benchmarking agent capabilities in ultra long-horizon scenarios, 2025. URL <https://arxiv.org/abs/2509.21766>.
- Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Belle-mare, Alex Graves, Martin Riedmiller, Andreas K. Fidjeland, Georg Ostrovski, Stig Petersen, Charles Beattie, Amir Sadik, Ioannis Antonoglou, Helen King, Dharmashan Kumaran, Daan Wierstra, Shane Legg, and Demis Hassabis. Human-level control through deep reinforcement learning. *Nature*, 518(7540):529–533, 2015.
- Junbo Niu, Yuanhong Zheng, Ziyang Miao, Hejun Dong, Chunjiang Ge, Hao Liang, Ma Lu, Bohan Zeng, Qiahao Zheng, Conghui He, and Wentao Zhang. Native visual understanding: Resolving resolution dilemmas in vision-language models, 2025.
- Mingyu Ouyang, Siyuan Hu, Kevin Qinghong Lin, Hwee Tou Ng, and Mike Zheng Shou. Game-World: Towards standardized and verifiable evaluation of multimodal game agents, 2026. URL <https://arxiv.org/abs/2604.07429>.
- Davide Paglieri, Bartłomiej Cupiał, Samuel Coward, Ulyana Piterbarg, Maciej Wołczyk, Akbir Khan, Eduardo Pignatelli, Łukasz Kuciński, Lerrel Pinto, Rob Fergus, et al. BALROG: Benchmarking agentic LLM and VLM reasoning on games. In *International Conference on Learning Representations*, volume 2025, pp. 96666–96702, 2025.
- Dongmin Park, Minkyu Kim, Beongjun Choi, Junhyuck Kim, Keon Lee, Jonghyun Lee, Inkyu Park, Byounguk Lee, Jaeyoung Hwang, Jaewoo Ahn, et al. Orak: A foundational benchmark for training and evaluating LLM agents on diverse video games. In *International Conference on Learning Representations*, volume 2026, pp. 60684–60744, 2026.

- Long Phan, Mantas Mazeika, Andy Zou, and Dan Hendrycks. TextQuests: How good are LLMs at text-based video games? *arXiv preprint arXiv:2507.23701*, 2025.
- Pooyan Rahmanzadehgervi, Logan Bolton, Mohammad Reza Taesiri, and Anh Totti Nguyen. Vision language models are blind: Failing to translate detailed visual features into words, 2024. URL <https://arxiv.org/abs/2407.06581>.
- scarsty and contributors. kys-cpp: a C++ reimplementation of 金庸群俠傳. GitHub repository, <https://github.com/scarsty/kys-cpp>, 2026.
- Jiaqi Shao, Hanck Chen, Wei Zhang, Maxm Pan, and Bing Luo. Do agent benchmarks measure capability? Protocol validity in the age of agentic AI, 2026. URL <https://arxiv.org/abs/2607.22368>.
- Zhi Rui Tam, Ya-Ting Pai, Yen-Wei Lee, and Yun-Nung Chen. VisTW: Benchmarking vision-language models for Traditional Chinese in Taiwan, 2025. URL <https://arxiv.org/abs/2503.10427>.
- Muhammad Usama, Syeda Aishah Asim, Syed Bilal Ali, Syed Talal Wasim, and Umair Bin Mansoor. Analysing the robustness of vision-language models to common corruptions, 2025.
- Jiayu Wang, Yifei Ming, Zhenmei Shi, Vibhav Vineet, Xin Wang, Yixuan Li, and Neel Joshi. Is a picture worth a thousand words? Delving into spatial reasoning for vision language models, 2024. URL <https://arxiv.org/abs/2406.14852>. NeurIPS 2024.
- Archana Warriar, Dat Nguyen, Michelangelo Naim, Moksh Jain, Yichao Liang, Karen Schroeder, Cambridge Yang, Joshua B Tenenbaum, Sebastian Vollmer, Kevin Ellis, et al. Benchmarking world-model learning with environment-level queries. *arXiv preprint arXiv:2510.19788*, 2025.
- Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh J Hua, Zhoujun Cheng, Dongchan Shin, Fangyu Lei, et al. OSWorld: Benchmarking multimodal agents for open-ended tasks in real computer environments. *Advances in Neural Information Processing Systems*, 37:52040–52094, 2024.
- Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ -bench: A benchmark for tool-agent-user interaction in real-world domains, 2024. URL <https://arxiv.org/abs/2406.12045>.
- Alex L Zhang, Thomas L Griffiths, Karthik R Narasimhan, and Ofir Press. VideoGameBench: Can vision-language models complete popular video games? *arXiv preprint arXiv:2505.18134*, 2025a.
- Haoran Zhang, Chenhao Zhu, Sicong Guo, Hanzhe Guo, Haiming Li, and Donglin Yu. StarBench: A turn-based RPG benchmark for agentic multimodal decision-making and information seeking, 2025b. URL <https://arxiv.org/abs/2510.18483>.
- Ruizhi Zhang, Ye Huang, Yuangang Pan, Chuanfu Shen, Zhilin Liu, Ting Xie, Haijun Lei, and Lixin Duan. Mastering PokeGym: Graph-guided multimodal evolution at test time, 2026a. URL <https://arxiv.org/abs/2604.08340>.
- Yunbei Zhang, Janet Wang, Yingqiang Ge, Weijie Xu, Jihun Hamm, and Chandan K. Reddy. Stop comparing LLM agents without disclosing the harness, 2026b. URL <https://arxiv.org/abs/2605.23950>.
- Yuxuan Zhu, Tengjun Jin, Yada Pruksachatkun, Andy Zhang, Shu Liu, Sasha Cui, Sayash Kapoor, Shayne Longpre, Kevin Meng, Rebecca Weiss, et al. Establishing best practices in building rigorous agentic benchmarks. *Advances in Neural Information Processing Systems*, 38, 2025.

A SESSION AND INFRASTRUCTURE PARAMETERS

A host of 8 vCPUs and 16 GiB runs up to 24 concurrent sessions. Emulation is DOSBox Pure through libretro at a fixed 26800 cycles, the speed of a 486DX2-66 and period-correct for the game, with no sound card, so the observation is visual only. The published video runs at 8 times speed or faster for long runs, with a strip above the frame naming the actor, the action number, the keys held and the clock, and a timeline file beside it carries every keypress and its hold duration for the replay.

B RELEASE AND LICENSING

The benchmark code is released under MIT. The emulation core (`dosbox_pure_libretro`) is GPLv2, loaded at runtime through the libretro C API and redistributed unmodified; the bundled macOS core is built from `schellingb/dosbox-pure` at `7f6e8fb`, and the hosted service ships a Linux build pinned by image digest. The game is the 1996 commercial release, copyright 智冠科技 and 河洛工作室. It is not redistributable, it is not tracked in the repository, and every host supplies its own copy. The public catalogue therefore carries only the metrics, videos and replay timelines we measured, and nothing that substitutes for the game data. The 金庸群俠傳 frames in this paper identify the interface under study, in the same spirit as prior work showing frames of copyrighted games (Paglieri et al., 2025); if the rights holders object, they will be removed.

C THE FOURTEEN BOOKS

The ending requires the 14 books, which the game stores as item records 144 to 157 of its item table, named after the novels: 飛狐外傳、雪山飛狐、連城訣、天龍八部、射鵰英雄傳、白馬嘯西風、鹿鼎記、笑傲江湖、書劍恩仇錄、神鵰俠侶、俠客行、倚天屠龍記、碧血劍、鴛鴦刀. A book counts as held when it is in the shared bag or in the four carried slots of the protagonist. With all fourteen in hand and sufficient reputation, the game opens its final sequence at a shrine on the far side of the map, where the books are placed and a last battle decides the ending (Bahamut forum contributors, 2023).

D THE FIELD AT SUBMISSION TIME

Table 4 lists the field, one row per model, with the number of sessions the model played at this budget. A model with more than one session is listed by the session that reached the most rungs, the most recent on a tie, and the committed records keep every session; a session marked *idle* ended after ten minutes without a key. The 3 sessions created under a variant spelling of the model name, `claude-fable-5.1-high` as `claude-fable5.1-high` and `claude-sonnet-5-high` as `claude-sonnet5-high` and `glm-5.3-flash-high` as `glm-5.3-flash`, are listed under the model, and the records keep the declared names. The live leaderboard at <https://hanxiao.io/jy-crpg-bench/> shows the catalogue as published and the runs added since.

Table 4: The field, one row per model: sessions played at this budget, minutes played by the reported session, how it ended, *time* at the budget or *idle* after ten minutes without a key, rungs reached, and the eight rungs, ✓ for reached and ○ for not reached, ordered by rungs reached and then by played time, with the random floor last.

Model	sessions	played	ended	rungs	item	map	world	compass	party	exp	lv 2	book
<code>gpt-6-astra</code>	4	60	time	4/8	✓	✓	✓	✓	○	○	○	○
<code>claude-opus5-high</code>	2	60	time	2/8	✓	✓	○	○	○	○	○	○
<code>gemini-3.8-flash</code>	4	60	time	2/8	✓	✓	○	○	○	○	○	○
<code>gpt-5.6-sol</code>	4	60	time	2/8	✓	✓	○	○	○	○	○	○
<code>grok-4.6</code>	2	60	time	2/8	✓	✓	○	○	○	○	○	○
<code>gemini-3.7-flash</code>	4	59	time	2/8	✓	✓	○	○	○	○	○	○
<code>gpt-5.6-terra</code>	6	46	time	2/8	✓	✓	○	○	○	○	○	○
<code>gpt-5.6-luna</code>	5	35	idle	2/8	✓	✓	○	○	○	○	○	○
<code>glm-5.3-flash</code>	5	60	time	1/8	○	✓	○	○	○	○	○	○
<code>gemini-3.6-flash</code>	2	60	time	1/8	○	✓	○	○	○	○	○	○
<code>claude-sonnet5-high</code>	2	60	time	1/8	○	✓	○	○	○	○	○	○
<code>claude-fable5.1-high</code>	2	57	time	1/8	○	✓	○	○	○	○	○	○
<code>random-baseline</code>	2	60	time	1/8	✓	○	○	○	○	○	○	○